

Logs with zeros? Some problems and solutions*

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Abstract

When studying an outcome Y that is weakly-positive but can equal zero (e.g. earnings), researchers frequently estimate an average treatment effect (ATE) for a “log-like” transformation that behaves like $\log(Y)$ for large Y but is defined at zero (e.g. $\log(1+Y)$, $\operatorname{arcsinh}(Y)$). We argue that ATEs for log-like transformations should not be interpreted as approximating percentage effects, since unlike a percentage, they depend on the units of the outcome. In fact, we show that if the treatment affects the extensive margin, one can obtain a treatment effect of any magnitude simply by re-scaling the units of Y before taking the log-like transformation. This arbitrary unit-dependence arises because an individual-level percentage effect is not well-defined for individuals whose outcome changes from zero to non-zero when receiving treatment, and the units of the outcome implicitly determine how much weight the ATE for a log-like transformation places on the extensive margin. We further establish a trilemma: when the outcome can equal zero, there is no treatment effect parameter that is an average of individual-level treatment effects, unit-invariant, and point-identified. We discuss several alternative approaches that may be sensible in settings with an intensive and extensive margin, including (i) expressing the ATE in levels as a percentage (e.g. using Poisson regression), (ii) explicitly calibrating the value placed on the intensive and extensive margins, and (iii) estimating separate effects for the two margins (e.g. using Lee bounds). We illustrate these approaches in three empirical applications.

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1 Introduction

When the outcome of interest Y is strictly positive, researchers often estimate an average treatment effect (ATE) in logs of the form $E_P[\log(Y(1)) - \log(Y(0))]$, which has the appealing feature that its units approximate percentage changes in the outcome.¹ A practical challenge in many economic settings, however, is that the outcome may sometimes equal zero, and thus the ATE in logs is not well-defined. When this is the case, it is common for researchers to estimate ATEs for alternative transformations of the outcome such as $\log(1+Y)$ or $\operatorname{arcsinh}(Y) = \log(\sqrt{1+Y^2} + Y)$, which behave similarly to $\log(Y)$ for large values of Y but are well-defined at zero. The treatment effects for these alternative transformations are typically interpreted like the ATE in logs, i.e. as (approximate) average percentage effects. For example, among the 11 papers published in the *American Economic Review* since 2018 that interpret a treatment effect for $\operatorname{arcsinh}(Y)$, all but one interpret the result as a percentage effect or elasticity.²

The main point of this paper is that identified ATEs that are well-defined with zero-valued outcomes should not be interpreted as percentage effects, at least if one imposes the logical requirement that a percentage effect does not depend on the baseline units in which the outcome is measured (e.g. dollars, cents, or yuan).

Our first main result shows that if $m(y)$ is a function that behaves like $\log(y)$ for large values of y but is defined at zero, then the ATE for $m(Y)$ will be *arbitrarily sensitive* to the units of Y . Specifically, we consider continuous, increasing functions $m(\cdot)$ that approximate $\log(y)$ for large values of y in the sense that $m(y)/\log(y) \rightarrow 1$ as $y \rightarrow \infty$. The common $\log(1+y)$ and $\operatorname{arcsinh}(y)$ transformations satisfy this property. We show that if the treatment affects the extensive margin (i.e. $P(Y(1) = 0) \neq P(Y(0) = 0)$), then one can obtain any magnitude for the ATE for $m(Y)$ by rescaling the outcome by some positive factor a . It is therefore inappropriate to interpret the ATE for $m(Y)$ as a percentage effect, since a percentage is inherently a unit-invariant quantity, while the ATE for $m(Y)$ depends arbitrarily on the units of Y .

The intuition for this result is that a “percentage” treatment effect is not well-defined for an individual for whom treatment increases their outcome from zero to a positive value. For example, in our application to Carranza et al. (2022) in Section 5, the treatment induces more people to have positive hours worked. The percentage change in hours is then not well-defined for individuals who would work positive hours under the treatment condition but zero hours under the control condition. Any average treatment effect that is well-defined with zero-valued outcomes must therefore implicitly assign a value for a change along the extensive margin. For logarithm-like transformations $m(\cdot)$, the importance of the extensive margin is determined implicitly by the units of Y . To see why this is the case, consider an individual who works positive hours only if they are treated, so that $Y(1) > 0$ and $Y(0) = 0$. Their treatment effect for the transformed outcome $m(Y)$ is $m(Y(1)) - m(0)$, which becomes larger if the units of Y are re-scaled by some $a > 1$, e.g. if we convert from weekly hours

¹That is, $\log(Y(1)/Y(0)) \approx \frac{Y(1)-Y(0)}{Y(0)}$ when $Y(1)/Y(0) \approx 1$.

²We found 17 papers overall using $\operatorname{arcsinh}(Y)$ as an outcome variable, of which 11 interpret the units; see Appendix Table 1.

worked to yearly hours worked. When the treatment has an extensive margin effect, the ATE for $m(Y)$ can thus be made large in magnitude by re-scaling Y by a large factor a . By contrast, if we re-scale Y by a small factor $a \approx 0$, such that the resulting outcomes are close to zero, then $m(Y) \approx m(0)$, and so the ATE for $m(Y)$ will be small. By varying the units of the outcome, we can thus obtain any magnitude for the ATE for $m(Y)$.

Our theoretical results also imply that if we re-scale the units of the outcome by a finite factor $a > 0$, the ATE for a log-like transformation $m(Y)$ will change by approximately $\log(a)$ times the effect of the treatment on the extensive margin. This result implies that sensitivity analyses that explore how the estimated ATE for $m(Y)$ changes with finite changes in the units of Y —or equivalently, how the ATE for $\log(c + Y)$ changes with the constant c —are essentially indirectly measuring the size of the treatment effect on the extensive margin.

We illustrate the practical importance of these results by systematically replicating recent papers published in the *American Economic Review* that estimate treatment effects for arcsinh-transformed outcomes. In line with our theoretical results, we find that treatment effect estimates using $\text{arcsinh}(Y)$ are sensitive to changes in the units of the outcome, particularly when the extensive margin effect is large. In half of the papers that we replicated, multiplying the original outcome by a factor of 100 (e.g. converting from dollars to cents) changes the estimated treatment effect by more than 100% of the original estimate. We obtain similar results using $\log(1 + Y)$ instead of $\text{arcsinh}(Y)$.

What, then, are alternative options in settings with zero-valued outcomes? Our second main result delineates the possibilities. We show that when there are zero-valued outcomes, there is no treatment effect parameter that satisfies all three of the following properties:

- (a) The parameter is an average of individual-level treatment effects, i.e. takes the form $\theta_g = E_P[g(Y(1), Y(0))]$, where g is increasing in $Y(1)$.
- (b) The parameter is invariant to re-scaling of the units of the outcome (i.e. $g(y_1, y_0) = g(ay_1, ay_0)$).
- (c) The parameter is point-identified from the marginal distributions of the potential outcomes.

This “trilemma” implies that any target parameter that is well-defined with zero-valued outcomes must necessarily jettison at least one of the three properties above. Of course, the choice of target parameter should depend on the economic question of interest. Which of the three properties the researcher prefers to forgo will thus generally depend on their context-specific motivation for using a log-like transformation in the first place.

To that end, [Section 4](#) highlights a menu of parameters that may be attractive depending on the researcher’s core motivation. We first consider the case where the researcher is interested in obtaining a causal parameter with an intuitive “percentage” interpretation. In this case, it may be natural to consider a parameter outside of the class of individual-level averages of the form $E_P[g(Y(1), Y(0))]$. One prominent option is $\theta_{\text{ATE}\%} = \frac{E[Y(1) - Y(0)]}{E[Y(0)]}$, the ATE in levels as a percentage of the baseline mean, which in many cases can be estimated via Poisson regression ([Santos Silva and Tenreiro](#),

2006; Wooldridge, 2010). The researcher might also consider alternative normalizations of the outcome that lead to intuitive units, e.g. expressing the outcome in per-capita units or converting it to a rank with respect to some reference distribution. Next, we suppose the researcher would like to capture concave preferences over the outcome; for example, the researcher might consider income gains to be more meaningful for individuals who are initially poor. In this case, it is natural to directly specify how much the researcher values a change along the extensive margin relative to the intensive margin—e.g., that a change from 0 to 1 is worth an x percent change along the intensive margin. Finally, suppose the researcher is interested in separately understanding the effects of the treatment along both the intensive and extensive margins. In this case, the researcher may target separate parameters for the two margins—e.g., $E[\log(Y(1)) - \log(Y(0)) \mid Y(1) > 0, Y(0) > 0]$, the average effect in logs for individuals with positive outcomes under both treatments, captures the intensive margin. Separate effects for the two margins are not generally point-identified, but can be bounded using the method in Lee (2009) or point-identified with additional assumptions (Zhang et al., 2008, 2009).

Section 5 provides a blueprint for estimating these alternative parameters in practice by applying our recommended approaches to three recent empirical applications, including a randomized controlled trial (RCT) (Carranza et al., 2022), a difference-in-differences (DiD) setting (Sequeira, 2016), and an instrumental variables (IV) setting (Berkouwer and Dean, 2022).

Related work. The use of log-like transformations for dealing with zero-valued outcomes has a long history. The use of the $\log(1 + Y)$ transformation dates to at least Williams (1937), while Bartlett (1947) considers both the $\log(1 + Y)$ and inverse hyperbolic sine transformations.³ More recent papers by Burbidge et al. (1988) and Bellemare and Wichman (2020), among others, provide results for $\text{arcsinh}(Y)$ that are frequently cited in economics papers using this transformation.⁴

Previous work has illustrated in simulations or selected empirical applications that results for particular transformations such as $\log(1 + Y)$ or $\text{arcsinh}(Y)$ may be sensitive to the units of the outcome (Aihounton and Henningsen, 2021; de Brauw and Herskowitz, 2021). In concurrent work, Mullahy and Norton (2023) show theoretically that the marginal effects from linear regressions using $\log(1 + Y)$ or $\text{arcsinh}(Y)$ are sensitive to the scaling of the outcome, with the limits of the marginal effects approaching those of either a levels regression or a (normalized) linear probability model, depending on whether the units are made small or large. We complement this work by proving that scale-dependence is a necessary feature of *any* identified ATE that is well-defined with zero-valued outcomes, and that the dependence on units is arbitrarily bad for transformations that approximate $\log(Y)$ for large values of Y . Thus, it is not possible to fix the issues with $\log(1 + Y)$ or $\text{arcsinh}(Y)$ by choosing a “better” transformation or using a different estimator. We also complement previous empirical examples by providing a systematic analysis of the sensitivity to scaling for papers in the *American Economic Review* using $\text{arcsinh}(Y)$.

³Bartlett (1947) proposes using $\text{arcsinh}(\sqrt{Y})$.

⁴MacKinnon and Magee (1990) propose transformations of the form $\text{arcsinh}(y\zeta)/\zeta$, where ζ is estimated by assuming $\text{arcsinh}(y\zeta)/\zeta$ is normally distributed conditional on covariates.